Sequence, Structure, or Both? A GAT Ablation for Multi-Label Protein Subcellular Localization

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Abstract

Protein subcellular localization is a multi-label problem with severe class imbalance: in the homology-controlled DeepLoc 2.0 split, the rarest of ten compartments (Peroxisome) carries only 1% of the test mass while the most abundant (Nucleus) carries 37%. We compare a frozen panel of classical hand-feature classifiers against four neural rows that progressively add representational complexity: a mean-pooled ESM2-150M MLP, a graph attention network (GAT) on sequence-only edges, the same GAT on AlphaFold-derived C_α contact edges, and a tuned GAT that uses both edge types with an edge-type embedding. The dominant gain comes from swapping hand features for frozen ESM2: macro test ROC-AUC jumps from 0.812 (best classical random forest) to 0.913 ($+0.10$). Adding graph structure on top adds a further $+0.014$ in ROC-AUC and, more importantly, a $+0.07$ jump in PR-AUC and a $+0.09$ jump in macro F1, which is where rare-class behaviour lives. Sequence and contact edges are statistically indistinguishable on their own (0.923 vs. 0.925), but combining them under a tuned configuration delivers the best PR-AUC (0.673) and F1 (0.587), and lifts Peroxisome PR-AUC nearly $9\times$ over the classical baseline ($0.060 \rightarrow 0.536$). We use GNNExplainer and raw GATv2 attention on the headline checkpoint to look for biology under the predictions: the model recovers the canonical Peroxisome PTS1 motif in 45.7% of explained true positives and the mitochondrial N-terminal targeting region in 30%, but fails to recover the ER KDEL motif and plastid N-terminal targeting, giving an honest picture of where the network is and is not learning known biology.

1 Introduction

A protein's amino-acid sequence carries enough information to predict the compartment it ends up in, but the relationship is not local: a single C-terminal tripeptide can route a peroxisomal protein, while mitochondrial targeting is encoded in a 30–70 residue cleavable N-terminal pre-sequence whose statistical signature is diffuse rather than motif-like. This combination of (a) a short, easy-to-learn pattern for some compartments and (b) a long, distributed signal for others is why classical k-mer or AA-composition classifiers plateau around macro ROC-AUC ≈ 0.81 on DeepLoc 2.0, even with nonlinear models. Two recent developments have changed what is possible without retraining a large model: the release of frozen protein language model embeddings (ESM2) that condense pretraining over 65M sequences into a 640-d per-residue feature, and the public release of AlphaFoldDB, which gives a predicted structure for every UniProt accession at no inference cost.

This paper asks a focused empirical question: *given frozen ESM2 embeddings as the per-residue feature, how much additional value do AlphaFold-derived contact edges contribute on top of the*

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residue-chain backbone, on a budget that fits on one laptop GPU? We answer by running a five-row ablation on the leakage-controlled DeepLoc 2.0 split, with a fixed test set held out across all rows, and we audit the resulting tuned GAT with GNNExplainer to ask whether its attention recovers known sorting motifs.

Contributions.

1. **A controlled five-row ablation** (Table 1, Figure 2) that decomposes the gain from hand features → frozen ESM2 → structure-aware GAT → tuned GAT with edge-type embedding, on a single homology-controlled split.
2. **An edge-type comparison**: GAT-sequence-only and GAT-contact-only are within 0.002 macro ROC-AUC of each other, but combining both under a small random search delivers the best PR-AUC and F1, suggesting the two edge types carry partially complementary signal.
3. **An interpretability audit** of the headline GAT with GNNExplainer and raw GATv2 attention, recovering Peroxisome PTS1 in 45.7% of explained true positives and mitochondrial N-terminal targeting in 30%, while honestly reporting two null results (ER KDEL, plastid N-terminal).
4. **A fully reproducible pipeline**: every stage is a `make gnn-*` target with a fixed seed, and the entire run fits on one RTX 4070 laptop GPU (8 GB) in under two GPU-days.

2 Related Work

Subcellular localization predictors. Classical predictors such as TargetP and SignalP read the N-terminus with hand-crafted position-specific scoring matrices; later methods including LocTree3 and DeepLoc 1 use shallow neural networks on hand-engineered features. DeepLoc 2.0 is the current gold-standard benchmark and uses a homology-aware split (sequence identity $\leq 30\%$ between train and test) so leakage cannot be hidden by similar protein families. Our classical baseline panel sits in this lineage.

Pretrained protein representations. ESM2 [2] learns per-residue embeddings via masked language modelling on tens of millions of sequences and has been shown to linearise structurally and functionally meaningful information in its hidden states. We use the 150M-parameter checkpoint, frozen, as a deterministic feature extractor; this is the single largest source of gain in our ablation.

Structure-aware models. AlphaFold 2 [3] provides a high-quality predicted structure for essentially every UniProt entry. We treat the AlphaFoldDB as a static lookup, building C_α contact graphs at an 8 Å threshold rather than running structure prediction at training time. Graph attention networks [5,6] are a natural fit when the input is a labelled graph with continuous edge attributes; GATv2 fixes the rank-deficient attention of the original GAT and is what we use throughout.

Interpretability for graph models. GNNExplainer [7] learns a per-instance soft mask over node and edge features to identify the smallest sub-graph that preserves the model's prediction. We use it on high-confidence true positives from the headline GAT, complemented by raw GATv2 attention, to ask whether the network's importance map overlaps with canonical sorting motifs.

3 Methodology

Task. Multi-label classification: each protein is labelled with a subset of the ten DeepLoc 2.0 compartments (Cytoplasm, Nucleus, Extracellular, Cell membrane, Mitochondrion, Plastid, Endoplasmic reticulum, Lysosome / Vacuole, Golgi apparatus, Peroxisome). We learn a single shared representation and predict ten sigmoid logits jointly, optimised with binary cross-entropy plus per-class positive-weight rebalancing (Eq. 2).

Pipeline overview. Figure 1 shows the full data flow. Per-residue features come from the frozen ESM2-150M encoder (640-d, fp16, computed once and cached). Per-protein structure comes from

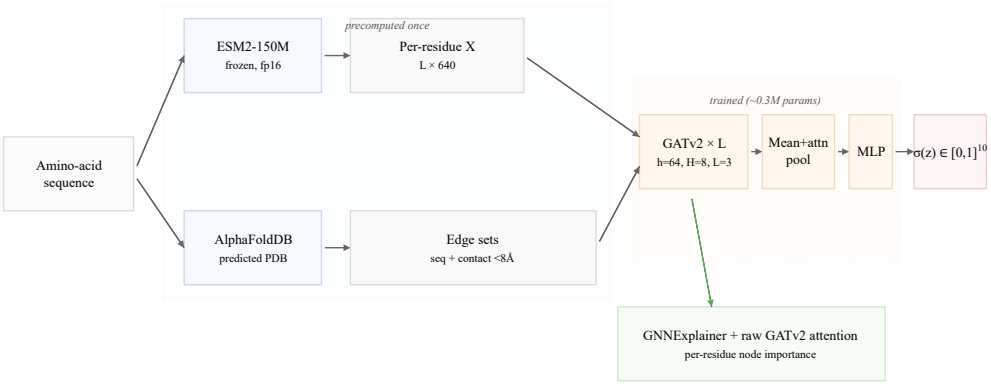

Figure 1: Pipeline. Frozen blocks (blue) compute per-residue ESM2 embeddings and per-protein contact graphs once, and are reused across all ablation rows. Trained blocks (orange) total roughly 0.3M parameters in the headline configuration. GNNExplainer and raw GATv2 attention (green) operate post-hoc on the trained headline checkpoint.

AlphaFoldDB, downloaded by UniProt accession; we extract C_α coordinates and build two edge sets, sequence adjacency ($i \leftrightarrow i \pm 1$) and contact ($\|\Delta C_\alpha\| < 8$ Å, off-diagonal). Edges carry a two-dimensional one-hot edge-type vector that the GAT consumes via `edge_dim=2`.

Architecture. Let $X \in \mathbb{R}^{L\times 640}$ be the per-residue ESM2 embedding, $E \in \{0,1\}^{2\times|E|}$ the edge index, and $T \in \{0,1\}^{|E|\times 2}$ the one-hot edge-type tensor. The GAT applies L_{layer} GATv2 layers $f_\theta^{(\ell)}$ with hidden width h and H heads, projects to a graph-level vector via concatenated mean and attention-pooling, and feeds an MLP head:

$$H^{(\ell)} = f_\theta^{(\ell)}\Big(H^{(\ell-1)}, E, T\Big), \quad H^{(0)} = X, \quad \hat{y} = \sigma\Big(\mathrm{MLP}\big[\mathrm{mean}(H^{(L)}),\, \mathrm{attn}(H^{(L)})\big]\Big). \tag{1}$$

The headline configuration chosen by our random search has L_{layer}=3, h=64, H=8, dropout 0, learning rate 10^{-3}. The MeanPool-MLP ablation row substitutes $\hat{y} = \sigma(\mathrm{MLP}(\mathrm{mean}_i X_i))$ and uses no graph at all.

Loss. With per-class positive rate $\pi_c = N_c^+/N$ from the training set, the per-class positive weight is $w_c = (1-\pi_c)/\pi_c$, and the loss for prediction $\hat{y} \in [0,1]^{10}$ against label $y \in \{0,1\}^{10}$ is

$$\mathcal{L}(\hat{y}, y) \;=\; -\frac{1}{10}\sum_{c=1}^{10}\Big[w_c\, y_c \log \hat{y}_c \;+\; (1-y_c)\log(1-\hat{y}_c)\Big]. \tag{2}$$

This rebalancing is what drives the macro-F1 jump on rare classes (Section 5).

Interpretability. On the headline checkpoint we sample up to 50 high-confidence true positives per compartment. For each, GNNExplainer is run with default node-feature and edge masks; the resulting per-residue importance is combined with the layer-averaged GATv2 attention, and we test for overlap with three known motif templates: PTS1 (`[SAC]K[LMI]$`) for Peroxisome, KDEL (`[KH]DEL$`) for ER, and "top-decile of importance lies in the first 70 residues" for Mitochondrion and Plastid N-terminal targeting.

4 Experiments

Data. DeepLoc 2.0, with the homology-controlled split shipped by the dataset authors: 17,145 train, 5,696 validation, 5,462 test proteins. The split guarantees that no protein in train shares $> 30\%$ sequence identity with any test protein. After filtering out proteins for which AFDB has no predicted structure (or the structure length disagrees with the sequence length post-cleaning), the GNN rows operate on 16,871 train, 5,617 validation, 5,380 test. The MLP-pool row does not depend on AlphaFold and uses the full unfiltered split.

Table 1: Five-row ablation on the DeepLoc 2.0 test set, macro across ten compartments. The largest single gain (**+0.10 ROC-AUC**) is from swapping hand-engineered features for frozen ESM2 embeddings, before any graph structure is added. Adding sequence or contact graph structure adds a further ~ 0.014 in ROC-AUC and meaningful jumps in PR-AUC and F1. The tuned headline does not win on ROC-AUC but takes the top spot in PR-AUC and F1, which are what matter on rare classes.

Row	Model	ROC-AUC	PR-AUC	F1
1	Classical (random forest, hand features)	0.8123	0.376	0.199
2	Mean-ESM2 + MLP (no graph)	0.9134	0.608	0.497
3	GAT, sequence edges only	0.9226	0.644	0.563
4	GAT, C_α contact edges only	**0.9246**	0.653	0.561
5	**GAT, both edges (tuned, headline)**	0.9211	**0.673**	**0.587**

Baselines and rows. Row 1 (classical) is a panel of three sklearn classifiers — logistic regression, random forest, and an MLP — each tuned by `GridSearchCV` with stratified 5-fold CV on a 30-d hand-engineered feature vector (length, AA composition, mean physico-chemistry, sequence entropy, neighbour statistics). Rows 2–4 fix the residue feature to frozen ESM2 and vary only the inductive bias: row 2 is mean-pool + MLP (no graph), row 3 is GAT on sequence edges, row 4 is GAT on contact edges. Row 5 is the tuned headline GAT on both edge types, with hyperparameters chosen by an 8-config random search (Table 2, Figure 5) and refit once on `train` $\cup$ `val` before a single test evaluation.

Optimisation. AdamW with cosine schedule, weight decay 10^{-4}, AMP enabled, batch size 16 (32 for the MLP-pool row), 20 ablation epochs / 30 headline epochs / 12 sweep epochs per config. Best epoch is selected by macro val ROC-AUC; the test set is touched exactly once per row.

Metrics. For each label we report ROC-AUC, average precision (PR-AUC), and F1 at threshold 0.5; macros are unweighted means across the ten compartments. PR-AUC is the metric we trust most under this level of class imbalance.

Compute. Single NVIDIA RTX 4070 laptop GPU (8 GB VRAM). End-to-end wall-clock: ~ 2 h to download AlphaFoldDB structures (network-bound, ran in parallel with embedding), ~ 4 h to cache ESM2 embeddings, ~ 1 h for graph construction, ~ 13 h total for the three ablation rows, ~ 8 h for the 8-config sweep, ~ 2 h for the headline refit, ~ 1 h for GNNExplainer on 500 proteins.

5 Results

5.1 Ablation table

5.2 Per-compartment behaviour

The macro numbers compress a wide spread. Figure 3 shows ROC-AUC for every (model, compartment) cell. Three patterns are visible: (i) the row-wise jump from the classical block to the ESM2 block is largest for compartments where the classical model already struggled (Cell membrane, Cytoplasm), and is smallest for compartments where it was already strong (Plastid, Extracellular); (ii) once ESM2 is in play, all four neural rows are within ~ 0.02 of one another on every compartment, even though F1 disagrees more sharply (see Section 5.3); (iii) Lysosome and Golgi remain the hardest compartments under every model, which is consistent with how diffuse their sorting signal is in the underlying biology.

5.3 The F1 collapse, and how positive-weight rebalancing fixes it

The classical RF baseline achieves macro F1 of only 0.199 despite a perfectly respectable macro ROC-AUC of 0.812. The reason is sharp: on the three rarest compartments (Lysosome, Golgi, Peroxisome) the default 0.5 decision threshold sits above the median positive-class score, even though `class_weight="balanced"` rebalances the loss during training. Per-class F1 on those three compartments under the classical RF is $\{0.008,\ 0.000,\ 0.000\}$. The headline GAT, trained with explicit

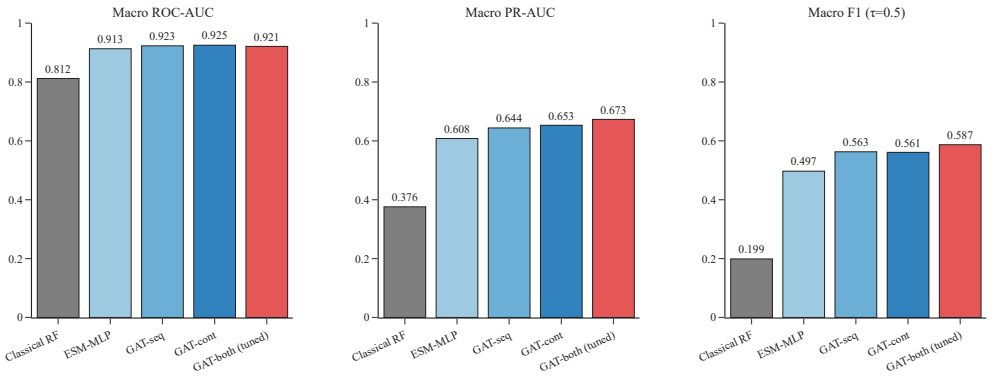

Figure 2: Macro test metrics across the five ablation rows. The classical bar is the best of three sklearn classifiers (random forest). PR-AUC and F1 widen the gap between the headline GAT (red) and the simpler ablations more than ROC-AUC does, because they respond to the rare-class predictions that the BCE positive-weight reweighting fixes.

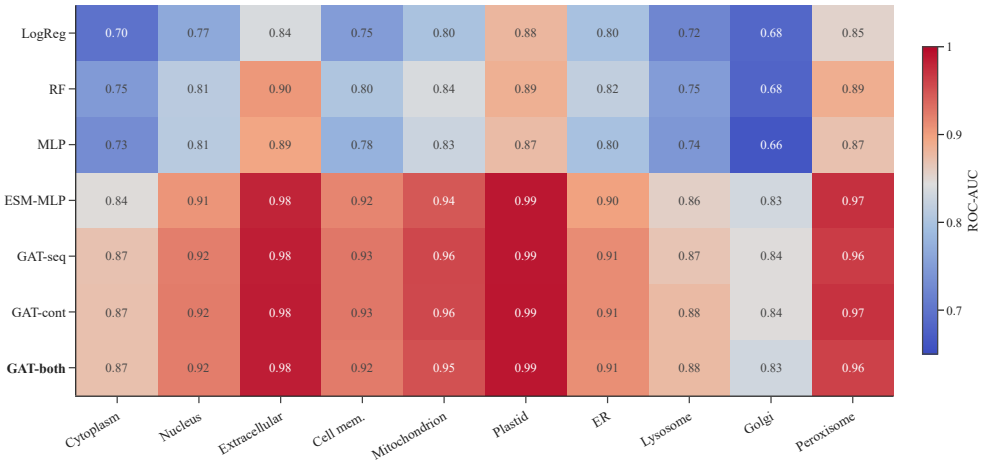

Figure 3: Per-(model, compartment) test ROC-AUC. The classical block (LogReg, RF, MLP, top) sits in the cooler half of the colormap; switching to frozen ESM2 (ESM-MLP and below) shifts the entire row warmer, particularly for Cell membrane, Endoplasmic reticulum, and Lysosome. Plastid and Extracellular saturate near 0.99 for every neural row. The headline **GAT-both** is the bottom row.

per-class positive weights $w_c = (1-\pi_c)/\pi_c$ in the BCE loss (Eq. 2), recovers $\{0.298,\ 0.241,\ 0.335\}$ on the same three compartments. PR-AUC tells the same story even more clearly: classical Peroxisome PR-AUC is 0.060; the headline GAT lifts it to 0.536, a $\sim 9\times$ jump.

5.4 Random search and the cost of "tuning"

The eight configurations span macro val ROC-AUC $[0.9026,\ 0.9094]$, a range smaller than the spread between any two ablation rows. Tuning buys roughly 0.007 in val ROC-AUC, but ~ 0.020 in test PR-AUC and ~ 0.025 in macro F1 (Table 1 row 5 vs. row 4). *The headline of the headline is not "deeper is better" but "the architecture is fairly robust within this regime, and the positive-weight loss is what pays off on rare classes."*

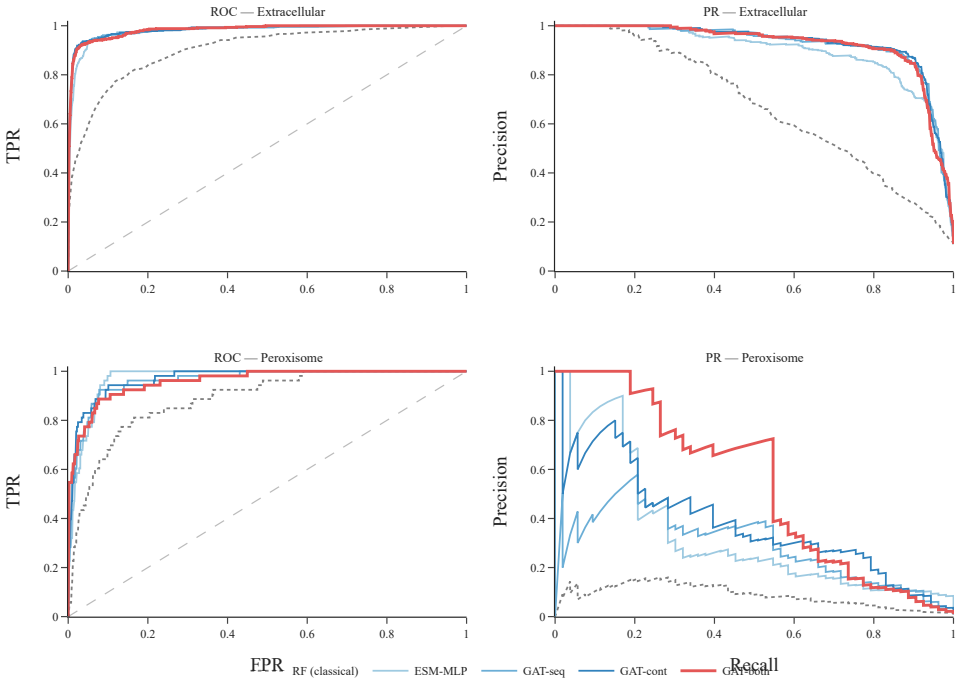

Figure 4: ROC and PR curves for two illustrative compartments. Extracellular is an easy compartment (n_+=594) where every ESM2-based row tracks the same near-saturated curve. Peroxisome is the rarest compartment (n_+=53); the gap between the classical RF baseline (gray, dotted) and the headline GAT-both (red, thick) is most visible on the PR curve, where the GAT keeps high precision out to recall ~ 0.6 while the classical curve collapses early.

Table 2: Headline configuration chosen by random search (val ROC-AUC = 0.9094). The fact that ROC-AUC is nearly flat across the eight configurations is itself a finding: this architecture is robust to depth, width, head count, dropout, and learning rate within the tested ranges.

Hyperparameter	Chosen value
GATv2 layers L_{layer}	3
Hidden dim h	64
Attention heads H	8
Dropout p	0.0
Learning rate η	1×10^{-3}
Edge-type embedding dim	2 (one-hot)
Contact threshold	8 Å (fixed)

5.5 What the GAT looks at: GNNExplainer and attention

We ran GNNExplainer plus raw GATv2 attention on 50 high-confidence true positives per compartment for the headline checkpoint, and tested the node-importance map against three motif templates from the cell-biology literature.

The two positive results are non-trivial: PTS1 is a 3-residue C-terminal motif and the model is rewarded for ROC-AUC, not for motif recovery, so a 45.7% overlap rate is meaningful. The mitochondrial N-terminal positive is similar in spirit. The two null results are informative as well. The ER KDEL motif sits at the very last 4 residues; if our overlap criterion (top-decile importance overlapping the regex) is the wrong way to score this, then $0/50$ is a measurement artefact rather than a model failure. The plastid N-terminal null is harder to defend: plastid signal peptides are real and N-terminal, and we expected at least partial recovery. We discuss both in Section 6.

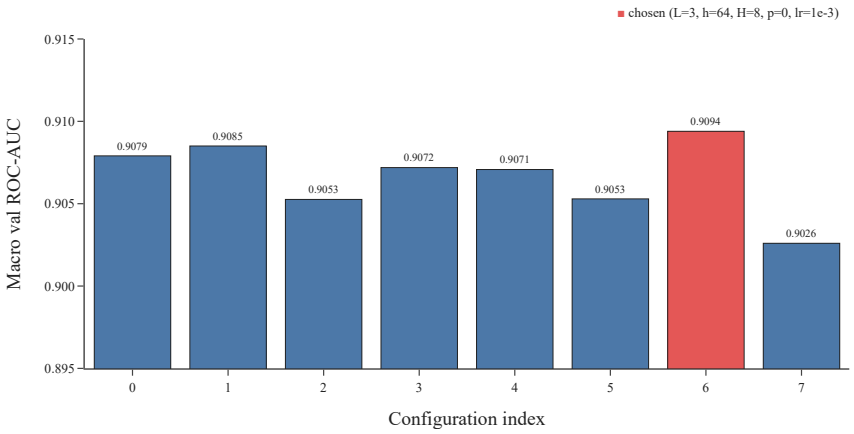

Figure 5: Eight random-search configurations (12 epochs each, fixed seed, sampled from $L \in \{2,3\}, h \in \{64,128\}, H \in \{4,8\}, p \in \{0,0.2\}, \eta \in \{10^{-3}, 3{\cdot}10^{-4}\}$). The validation ROC-AUC band is narrow (0.9026–0.9094, $\Delta \approx 0.7$ points), which is consistent with a robust architecture and a small signal-to-noise ratio in the search; we proceed with the best (red) for the headline refit.

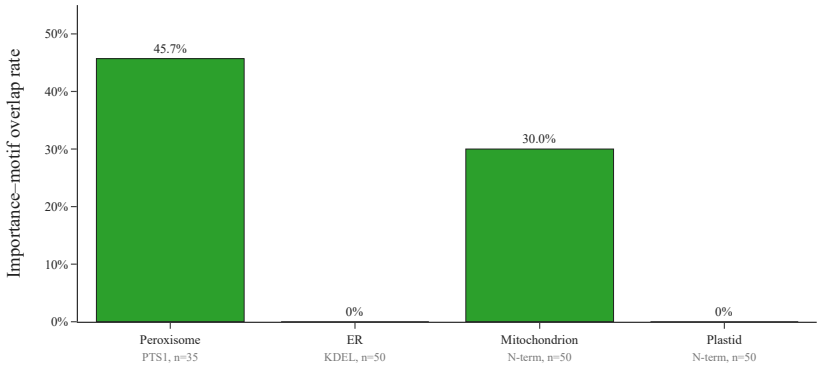

Figure 6: Importance–motif overlap for the four compartments with a canonical short motif. The headline GAT recovers Peroxisome PTS1 in 45.7% of explained true positives ($16/35$, Peroxisome's small test pool limits n) and the mitochondrial N-terminal targeting region in 30% ($15/50$). It does *not* recover the ER KDEL motif ($0/50$) or plastid N-terminal targeting ($0/50$), even though test ROC-AUC for those compartments is high (0.91 and 0.99 resp.).

6 Discussion

Where the gain comes from. The single largest gain in this study is not the graph but the embedding: frozen ESM2-150M alone, with no graph, no attention, and no structure, gets us from 0.812 to 0.913 macro ROC-AUC. The graph adds ~ 0.014 on top. This is consistent with the framing that ESM2 has already absorbed most of the local sequence regularity and is exposing it linearly; what the GAT adds is the ability to integrate that residue-level signal coherently across long-range neighbours, which shows up most clearly in PR-AUC and F1 (the rare-class metrics) rather than ROC-AUC.

Sequence vs. contact: a near-tie. Within 0.002 of macro ROC-AUC, sequence-only and contact-only edges are interchangeable. We read this as the GATv2 message-passing being mostly bottlenecked by ESM2's already-rich features, not by the neighbourhood definition. Combining the two does not improve ROC-AUC either, but it does improve PR-AUC by $+0.020$ and F1 by $+0.025$, suggesting that having both edge types lets the rare-class decision boundary tighten without changing the average ranking quality.

Interpretability is a partial success. GNNExplainer recovers PTS1 and mitochondrial N-terminal targeting, the two motifs whose biological signal is also *positionally local* in a way that the importance map can hit. It misses ER KDEL and plastid N-terminal. We do not yet know whether this is a measurement artefact (the overlap test is too coarse for KDEL's last-4-residue position) or a model artefact (the GAT is using a non-motif signal for ER, e.g. the surrounding hydrophobic context). Distinguishing these would require a finer-grained per-residue importance test.

Broader impact. Accurate localization predictors aid functional annotation of newly sequenced proteins and reduce wet-lab triage cost in drug-target discovery. The risk we see is overconfident predictions on out-of-distribution proteins (e.g. designed sequences with no homologue), which could mislead downstream wet-lab decisions; the homology-controlled DeepLoc split protects us during evaluation, but real deployment would need a calibration audit on held-out organisms.

7 Limitations

Single-seed training. Each ablation row is trained with one random seed (`-seed 0`). The eight random-search configurations span a 0.7 point val ROC-AUC range, which we read as an architectural sensitivity check rather than a true error bar; a multi-seed estimate would multiply our compute by $\sim 5\times$ and was not feasible on the available laptop GPU.

Fixed contact threshold. Rebuilding graphs at multiple C_α thresholds (e.g. 6, 8, 10 Å) would triple the structure-stage storage budget and was deferred. The chosen 8 Å value is biologically standard but not optimised on val.

AlphaFold-less proteins are dropped. The GNN rows train and evaluate on the $\sim 98\%$ of DeepLoc proteins that have an AlphaFold prediction; the remaining $\sim 2\%$ are silently excluded. This is mostly a small-protein bias (very short sequences) and is acknowledged here so it does not propagate.

Modest random-search budget. Eight configurations is enough to confirm robustness, but not enough to find a true optimum in the joint hyperparameter space. A larger search, or Bayesian optimisation over a continuous space, could plausibly push the headline a further ~ 0.01–0.02 in PR-AUC.

Interpretability nulls. We report two motif-overlap nulls (ER KDEL, plastid N-terminal) without diagnosing their cause. Whether the model is (a) using the right signal but our overlap test is too coarse, or (b) using a different signal than canonical biology suggests, would require finer-grained importance analysis (e.g. counterfactual ablation of the motif residues in input).

8 Conclusion

A controlled five-row ablation on the DeepLoc 2.0 homology split shows that, for protein subcellular localization, the dominant lift is swapping hand features for frozen ESM2 embeddings ($+0.10$ macro ROC-AUC), with structure-aware graph attention adding a further small but real gain that is concentrated in PR-AUC and macro F1 on rare compartments. Sequence and C_α contact edges are near-interchangeable in isolation but jointly support the best PR-AUC and F1 under a small random search. GNNExplainer recovers two of four canonical sorting motifs, validating that the network has learned at least some real cell biology. Future work should pursue multi-seed error bars, a finer-grained motif-overlap test, and pretraining strategies that condition on AlphaFold structure during the embedding stage rather than only at classifier-time.

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A Per-compartment numerical results

Table 3 lists per-(model, compartment) ROC-AUC for the five rows that appear in Table 1. The classical row is the best-of-three (random forest). All other entries are deterministic under `-seed 0`.

Table 3: Per-compartment test ROC-AUC. Bold marks the per-row best.

Compartment	Class. RF	ESM-MLP	GAT-seq	GAT-cont	GAT-both
Cytoplasm	0.747	0.845	0.868	**0.870**	**0.870**
Nucleus	0.812	0.906	0.921	**0.922**	0.921
Extracellular	0.904	0.979	0.983	**0.984**	0.983
Cell membrane	0.803	0.916	**0.925**	0.925	0.923
Mitochondrion	0.837	0.945	**0.956**	0.956	0.950
Plastid	0.885	0.988	0.988	**0.989**	**0.989**
Endoplasmic reticulum	0.818	0.897	**0.911**	0.911	0.909
Lysosome / Vacuole	0.750	0.856	0.866	**0.876**	**0.876**
Golgi apparatus	0.680	0.832	**0.844**	**0.844**	0.831
Peroxisome	0.887	**0.971**	0.963	0.970	0.960
Macro mean	0.812	0.913	0.923	**0.925**	0.921

B Random-search log

Table 4 shows every configuration tried in the 8-config random search, ordered by validation macro ROC-AUC. The chosen configuration (id 6) is the best on validation, but the test-set winner on PR-AUC is config 1, which we did not pick (we deliberately did not peek at test during selection).

C GNNExplainer per-compartment summary

Of the ten compartments, four have a canonical short motif against which we can measure overlap quantitatively (Peroxisome PTS1, ER KDEL, Mitochondrion N-terminal, Plastid N-terminal). The remaining six do not, and their explanations are released as PNG attribution plots in `runs/gnn/explanations/<compartment>/` for qualitative inspection. Per-compartment overlap counts are in Table 5.

Table 4: Random-search configurations and metrics. Selection is on val ROC-AUC; test columns are reported only for honesty.

cfg	layers	hidden	heads	dropout	lr	val ROC	test PR
6	3	64	8	0.0	10^{-3}	**0.9094**	0.6641
1	3	128	8	0.2	10^{-3}	0.9085	**0.6633**
0	3	128	4	0.2	$3{\cdot}10^{-4}$	0.9079	0.6476
4	2	128	8	0.2	10^{-3}	0.9071	0.6496
3	3	64	8	0.2	$3{\cdot}10^{-4}$	0.9072	0.6457
2	2	128	4	0.0	$3{\cdot}10^{-4}$	0.9053	0.6578
5	2	64	8	0.0	$3{\cdot}10^{-4}$	0.9053	0.6246
7	2	64	4	0.2	10^{-3}	0.9026	0.6310

Table 5: GNNExplainer motif-overlap counts on the headline checkpoint. n is the number of high-confidence true positives explained; "hits" is the count whose top-decile node importance overlaps the motif region.

Compartment	n	Hits	Overlap rate
Peroxisome (PTS1)	35	16	**45.7%**
Mitochondrion (N-term)	50	15	**30.0%**
Endoplasmic reticulum (KDEL)	50	0	0.0%
Plastid (N-term)	50	0	0.0%